

Two Stacks Sorting

Time limit: 1.00 s

Memory limit: 512 MB

You are given an input list that consists of n numbers. Each integer between 1 and n appears exactly once in the list.

Your task is to create a sorted output list using two stacks. On each move you can do one of the following:

- Move the first number from the input list to a stack
- Move a number from a stack to the end of the output list

Input

The first input line has an integer n .

The second line has n integers: the contents of the input list.

Output

Print n integers: for each number the stack where it is moved (1 or 2).

You can print any valid solution. If there are no solutions, print IMPOSSIBLE.

Constraints

- $1 \leq n \leq 2 \cdot 10^5$

Example

Input	Output
5 2 3 1 5 4	1 2 1 1 2